SEAL: Interactive Tool for Systematic Error Analysis and Labeling

Nazneen Rajani[†], **Weixin Liang**[‡], **Lingjiao Chen**[‡], **Meg Mitchell**[†], **James Zou**[‡]
[†] Hugging Face [‡] Department of Computer Science, Stanford University
{nazneen, meg}@huggingface.co {wxliang,lingjiao,jamesz}@stanford.edu

arXiv:2210.05839v1 [cs.CL] 11 Oct 2022

Abstract

With the advent of Transformers, large language models (LLMs) have saturated well-known NLP benchmarks and leaderboards with high aggregate performance. However, many times these models systematically fail on tail data or rare groups not obvious in aggregate evaluation. Identifying such problematic data groups is even more challenging when there are no explicit labels (e.g., ethnicity, gender, etc.) and further compounded for NLP datasets due to the lack of visual features to characterize failure modes (e.g., Asian males, animals indoors, waterbirds on land etc.). This paper introduces an interactive Systematic Error Analysis and Labeling (SEAL) tool that uses a two-step approach to first identify high error slices of data and then in the second step introduce methods to give human-understandable semantics to those under-performing slices. We explore a variety of methods for coming up with coherent semantics for the error groups using language models for semantic labeling and a text-to-image model for generating visual features. SEAL toolkit and demo screencast is available at https://huggingface.co/spaces/nazneen/seal.

1 Introduction

Machine learning systems that seemingly perform well on average can still make *systematic errors* on important subsets of data. Examples include such systems performing poorly for marginalized groups in chatbots (Stuart-Ulin, 2018), recruiting tools (Hamilton, 2018), cloud products (Kayser-Bril, 2020), ad targeting (Hao, 2019), credit services (Knight, 2019), and image cropping (Hamilton, 2020). Discovering and labeling systematic errors in ML systems is an open research problem that would enable building robust models that generalize across subpopulations of data.

Uncovering underperforming groups of data of a ML system is not straightforward. Firstly, the high-dimensional space of the representations learned by the deep learning models makes it difficult to identify such groups of systematic errors. Secondly, it is difficult to extract and label the hidden semantic information in such groups with high errors without a human-in-the-loop setup. Identifying systematic model failures requires practitioners to think creatively about model evaluation (Ribeiro et al., 2020; Wu et al., 2019; Goel et al., 2021b; Kiela et al., 2021; Yuan et al., 2022). However, current approaches are mostly limited to examining and manipulating model mispredictions. The onus of identifying what group or subset of data to evaluate still falls on the practitioner, making it inefficient and prone to oversight. Recent works on fine-grained error analysis, such as Domino (Eyuboglu et al., 2022) and Spotlight (d'Eon et al., 2022) provide solutions to this problem but focus on image datasets which are easier to visualize.

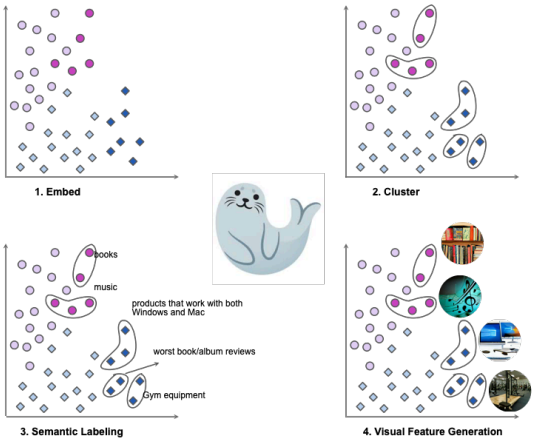

Figure 1: SEAL interactive tool for discovering systematic errors in model performance. Steps 1 and 2 include extracting the model embeddings and clustering datapoints with high-loss. Steps 3 and 4 include semantic labeling of error groups and generating visual features to support debugging.

Error analysis for text data is less explored and more challenging. It also highlights the need to provide semantic summaries of text, which we

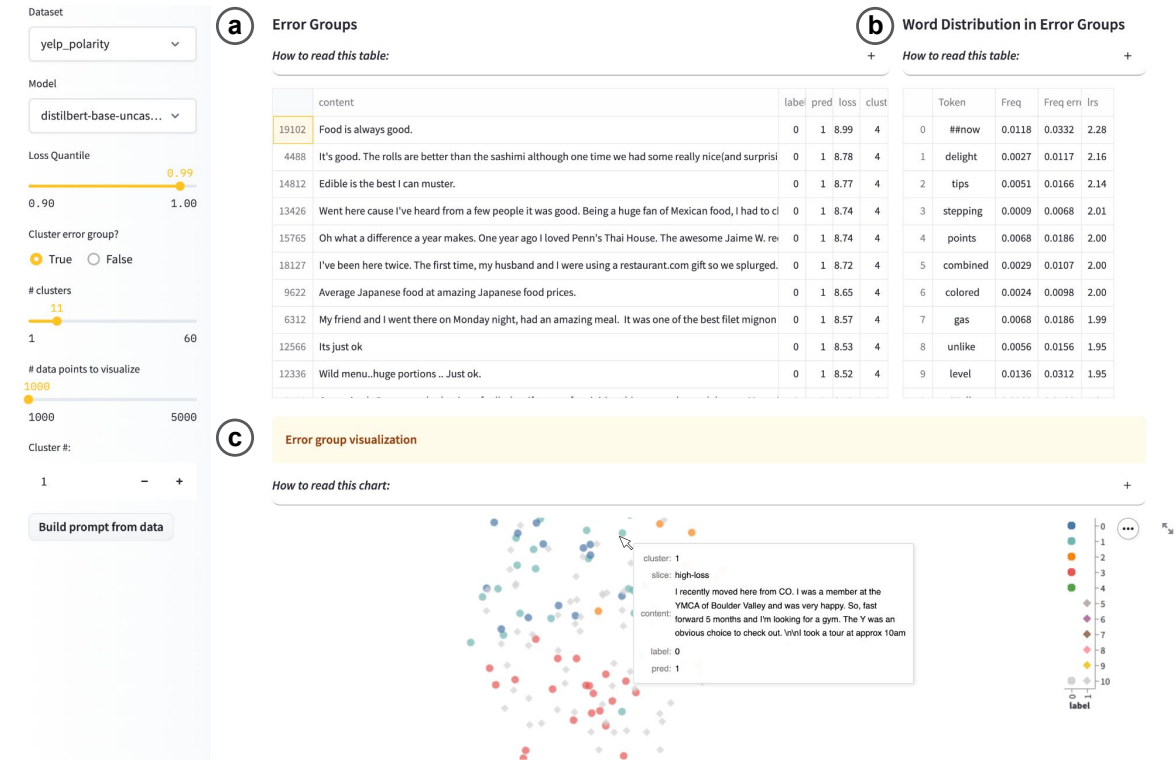

Figure 2: SEAL interface showing high-error groups for the distilbert-base-uncased model evaluated on the yelp_polarity dataset. The interface comprises of various components: (a) examples from the dataset in the high error groups (sorted by loss), (b) statistics of tokens in high error groups relative to the entire evaluation set, (c) interactive 2d visualization of the model embeddings showing groups of errors in color and low-loss groups in gray. The colors indicate different error clusters. If the dataset has annotated classes, the visualization includes symbols to represents those classes ($\diamond$ and $\circ$ in the above figure). The panel on the left has multiple widgets that a user can control to be able to interactively understand their model's mispredictions relative to the rest of the model's outputs. Apart from the dataset and model, the user can select the loss quantile that want to examine for systematic errors, if they want SEAL to group those errors using kmeans++ with the number of clusters, and how many data points they want to visualize at a time in the visual component of the interface downsampled proportional to the group size (we use Altair for plotting that supports a maximum of 5000 data points to be visualized at once).

tackle in SEAL. For example, NLP models could underperform on hundreds of possible input types – longer inputs, inputs from non-native speaker, inputs with topic domains underrepresented in training, etc. This is a huge barrier of entry for most non-expert ML users who wish to gain a better understanding of their model and datasets with such existing tools. Model evaluation should ideally give actionable insights into a model's performance on a dataset in the form of data curation (Liang and Zou, 2022) or model patching (Goel et al., 2021a).

Our desiderata is a tool that summarizes failures of a model on textual data in a concise, coherent and human intepretable way. Systematic Error Analysis and Labeling (SEAL) is an interactive tool to 1. identify candidate groups of data with high systematic errors and 2. generate semantic labels for those groups. For 1, we use k-means++ on subset of evaluation data with highest loss. Semantic labeling uses LLMs (like GPT3) in zero-shot setting for identifying concepts or topics common to examples in the candidate group. We also explored using a text-to-image model to generate visual features for high error clusters using the Dall-e-mini (Dayma et al., 2021). Semantic descriptions (via labeling or visual features) of such systematic model errors not only enable practitioners to better understand the failure modes of their model during evaluation but also gives actionable insight to fix them via some form of model patching or data augmentation.

2 SEAL

We present Systematic Error Analysis and Labeling (SEAL), an interactive visualization tool that provides rich data point comparison for text classification systems, enabling fine-grained understanding

of model performance on data groups as shown in Figure 2. It comes pre-loaded with model outputs for most downloaded HuggingFace (HF) models and datasets, as well as scripts for loading data for any dataset provided by the Datasets API and extracting embeddings of any HF-compatible model.[1]

2.1 Error Discovery and Analysis

Identifying model failures via error discovery is a crucial step in engineering robust systems that generalize to diverse subsets of data. SEAL uses the model's loss on a datapoint as a proxy for potential bugs or errors. Past work has examined model behavior on individual datapoints for mapping training datasets (Swayamdipta et al., 2020). We hope to leverage information about model behavior on individual *evaluation* data-points in a similar fashion. We use quantiles for dividing the model loss region for further analysis. For example, Figure 2 shows the 0.99 loss quantile for the distilbert-base-uncased model (Sanh et al., 2019) on the yelp_polarity (Zhang et al., 2015) sentiment classification dataset. The SEAL interface allows the user to control the loss quantile for fine-grained analysis using the widget on the side panel.

SEAL uses k-means++ for clustering the high-loss candidate datapoints from the above step. Meng et al. (2022) used k-means for topic discovery on the entire dataset and showed that the clusters are stable only when k is very high ($k >> 100$) because of the scale of the embedding space. In contrast, SEAL only clusters the very high loss slice (> 0.98 quantile).

We use the representations of the models' final hidden layer (before the softmax) as embeddings. If the evaluation dataset selected by the user has ground truth annotations, then it groups the clusters by error-types (false-positives and false-negatives for binary classification). The visualization component of the SEAL interface shows the error clusters and their types using colors and symbols respectively. We use a standard heuristic of setting the number of clusters in k-means++ to be approximately $\sqrt{n/2}$, where n is the group size.

2.2 Semantic Error Labeling

Semantic error labeling is important for identifying the underlying concept or topic connecting the datapoints in a error group. Systematic errors can be mathematically modeled and fixed by data curation. Contrast this with random errors that cannot be mathematically modeled or fixed via data curation. Past work analyzing NLP models have shown systematic errors on various tasks including sentiment classification, natural language inference, and reading comprehension (McCoy et al., 2019; Kaushik et al., 2020; Jia and Liang, 2017). SEAL uses pretrained LLMs (such as GPT3 (Ouyang et al., 2022) or Bloom (BigScience, 2022)) for semantic labeling of error clusters that could highlight such possible systematic bugs in model performance. We craft a prompt consisting of instruction and examples in the clusters extracted in the previous step as follows.

```
def build_prompt(content)
    instruction = 'In this task, we`ll assign a short and precise label to a group of documents based on the topics or concepts most relevant to these documents. The documents are all subsets of a ${task} dataset.'

    examples = '\n - '.join(content)

    prompt = instruction + '- ' + examples+ '\n Group label:'

    return prompt
```

Here `task` is the task under consideration for example 'sentiment classification' in our case. The `arg` to the function is a dataframe or dataframe column with the dataset content as string that the model uses for classification. Our prompt design was experimented first in the few-shot setting before adapting to the zero-shot.

For the results and use case discussion in Section 3, we use the OpenAI GPT3 API [2] via the CLI. The maximum token length is limited to 4000 and so we truncate the prompt to that length before feeding the model. We observed that for many larger groups of high-loss examples (> 25) SEAL labels degenerate to generic output such as "customer reviews of products", "movies reviews", "restaurant reviews", etc. To prevent this and to generate coherent group labels, we sub-cluster the bigger error groups until their size is < 25. We verified the group labels by running the Blei et al. (2003) LDA topic model on the examples in each cluster after a pre-processing step. The pre-processing included tokenizing, lemmatizing, and removing stopwords. For each dataset domain, we also removed the domain word list – ('movie, watch, film, character'

[1] Based on usage data from July'22 at https://huggingface.co/models?pipeline_tag=text-classification&sort=downloads

[2] https://beta.openai.com/playground